

Variational Monte Carlo for the Bose–Hubbard Model

Code Documentation

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1 Overview

This document describes a variational Monte Carlo (VMC) solver for the Bose–Hubbard (BH) model on periodic 1D and 2D lattices. The code estimates the ground-state energy E_0 by stochastically sampling a parameterized trial wavefunction Ψ_T using the Metropolis–Hastings algorithm.

2 Model Hamiltonian

The Bose–Hubbard Hamiltonian is

$$\hat{H}_{\text{BH}} = -t \sum_{\langle i,j \rangle} \left(\hat{b}_i^\dagger \hat{b}_j + \text{h.c.} \right) + \frac{U}{2} \sum_i \hat{n}_i (\hat{n}_i - 1) - \mu \sum_i \hat{n}_i, \quad (1)$$

where t is the hopping amplitude, U is the on-site interaction strength, μ is the chemical potential, $\hat{b}_i^{(\dagger)}$ are the bosonic annihilation (creation) operators, and $\hat{n}_i = \hat{b}_i^\dagger \hat{b}_i$ is the number operator on site i . For our purposes, we ignore the chemical potential since move proposals never

3 Code Parameters

Below are all the parameters which you can control when running this VMC code. Most are self-explanatory, but I will take some time to explain the not-so-obvious parameters; specifically, I will discuss the parameters under "Optimization MC" and "Final MC." We distinguish between these two types of Monte Carlo as we perform shorter Monte Carlo simulations (Optimization MC) to measure the energy and energy gradient for gradient descent to obtain the optimal parameters, then use these optimized parameters in a longer Monte Carlo simulation (Final MC) to obtain precise ground state energy measurements.

num-walkers is the number of separate system configurations/Markov chains that we perform Monte Carlo moves on. **num-MC-steps** is the number of Monte Carlo moves the code attempts to make on the walkers/configurations. **num-equil-steps** is the number of Monte Carlo moves the code attempts to make before making measurements to ensure the system is equilibrated. **block-size** is the size of each block of averaged measurements in the Monte Carlo simulations.

Table 1: Command-line options for the BH-VMC code.

Flag	Type	Default
<i>Physical parameters</i>		
-D, --dimension	Int	required (1 or 2)
-L, --length	Int	required for 1D
--length-x	Int	required for 2D
--length-y	Int	required for 2D
-N, --particle-number	Int64	required
-U, --interaction	Float64	required
-t, --t	Float64	1.0
<i>Trial wavefunction</i>		
--trial-state	String	required
--kappa	Float64	1.0
--jastrow-potentials	String	" "
--n-max	Int64	N
<i>Stochastic reconfiguration</i>		
--eta	Float64	0.01
--seed	Int64	1234
<i>Optimization MC</i>		
--no-optimize	flag	false
--max-iters	Int64	100
--opt-num-walkers	Int64	100
--opt-num-MC-steps	Int64	5000
--opt-num-equil-steps	Int64	1000
--opt-block-size	Int64	500
<i>Final MC</i>		
--final-num-walkers	Int64	200
--final-num-MC-steps	Int64	10000
--final-num-equil-steps	Int64	2000
--final-block-size	Int64	1000
<i>Output and miscellaneous</i>		
--output-dir	String	"data/VMC"
--save-history	flag	false
--skip-timing	flag	false
-h, --help	flag	—

4 Observables

In exact diagonalization, we must compute the full energy expectation value:

$$E = \langle H \rangle = \frac{\langle \psi_T | H | \psi_T \rangle}{\langle \psi_T | \psi_T \rangle}, \quad (2)$$

which is a sum over the entire Hilbert space for our system. Instead of brute force calculating the full expectation value, we can manipulate this expression to obtain a Monte Carlo-friendly form. The manipulation goes as follows:

$$\begin{aligned}
E\langle\psi_T|\psi_T\rangle &= \langle\psi_T|H|\psi_T\rangle \\
&= \sum_n \langle\psi_T|n\rangle\langle n|H|\psi_T\rangle \\
&= \sum_n (\langle\psi_T|n\rangle\langle n|H|\psi_T\rangle) \left(\frac{\langle n|\psi_T\rangle}{\langle n|\psi_T\rangle} \right) \\
&= \sum_n \frac{\langle n|H|\psi_T\rangle}{\langle n|\psi_T\rangle} |\langle n|\psi_T\rangle|^2,
\end{aligned}$$

where we have defined a quantity named the local energy as

$$E_{\text{loc}}(n) = \frac{\langle n|H|\psi_T\rangle}{\langle n|\psi_T\rangle} \quad (3)$$

and the probability as

$$p(n) = |\langle n|\psi_T\rangle|^2. \quad (4)$$

Assuming normalization, the Monte Carlo-friendly total energy expectation value is then

$$E = \sum_n p(n) E_{\text{loc}}(n), \quad (5)$$

where we have assumed normalization of the trial wavefunction: i.e., $\langle\psi_T|\psi_T\rangle = 1$. We can further expand the local energy in order to calculate it for each time we sample in the Monte Carlo simulation:

$$E_{\text{loc}}(\vec{n}) = \frac{\langle\vec{n}|H|\psi_T\rangle}{\langle\vec{n}|\psi_T\rangle} = \sum_{\vec{n}'} \frac{\langle\vec{n}|H|\vec{n}'\rangle\psi_T(\vec{n}')}{\psi_T(\vec{n})}, \quad (6)$$

which only sums over the system configurations $\vec{n}'$ that are connected to $\vec{n}$ by the Hamiltonian ($\langle\vec{n}|H|\vec{n}'\rangle$) weighted by $\psi_T(\vec{n}')/\psi_T(\vec{n})$.

5 Trial Wavefunction

Before detailing the two trial wavefunctions implemented in this code, we must define the arbitrary Fock state that mathematically describes any system configuration. For a system with M sites, the Fock state is

$$|\vec{n}\rangle = |n_1, n_2, \dots, n_M\rangle = \prod_{i=1}^M \frac{(b_i^\dagger)^{n_i}}{\sqrt{n_i!}} |0\rangle. \quad (7)$$

Additionally, we can define the diagonal observables, which are independent of the trial wavefunction:

5.1 Local Potential Energy

The local potential energy is calculated as follows:

$$V_{\text{loc}}^i = \frac{\langle n|V|\psi_T\rangle}{\langle n|\psi_T\rangle} = \sum_{n'} \langle n| \frac{U}{2} n_i (n_i - 1) |n'\rangle \frac{\langle n'|\psi_T\rangle}{\langle n|\psi_T\rangle} \quad (8)$$

$$V_{\text{loc}}^i = \frac{U}{2} n_i (n_i - 1). \quad (9)$$

5.2 Local Density-Density Correlation Function

The local density-density correlation function is calculated as follows:

$$C_{\text{loc}}^{i,j} = \frac{\langle n|n_i n_j|\psi_T\rangle}{\langle n|\psi_T\rangle} = \sum_{n'} \langle n|n_i n_j|n'\rangle \frac{\langle n'|\psi_T\rangle}{\langle n|\psi_T\rangle} \quad (10)$$

$$C_{\text{loc}}^{i,j} = n_i n_j. \quad (11)$$

For a translationally invariant periodic lattice, correlations are reported for each nonzero Euclidean distance shell $q = 2, \dots, Q$ as

$$C_q = \frac{1}{N_q} \sum_{\substack{i,j=1 \\ s(i,j)=q}}^M n_i n_j, \quad (12)$$

where N_q is the number of ordered site pairs in shell q . The on-site shell is excluded. In 1D this normalization reproduces the average over sites at each periodic separation.

5.3 Gutzwiller

The Gutzwiller trial wavefunction is defined as

$$|\psi_G\rangle = \prod_j \left(\sum_{n_j=0}^{n_{\text{max}}} f_{n_j} \frac{(b_j^\dagger)^{n_j}}{\sqrt{n_j!}} \right) |0\rangle_j, \quad (13)$$

where the variational weights are parameterized as

$$f_{n_j} = \mathcal{N} e^{-\kappa n_j^2}, \quad (14)$$

with $\mathcal{N}$ a normalization constant and $\kappa > 0$ a single variational parameter. One thing you should notice about the Gutzwiller trial wavefunction is that it is a product state of separate site states: i.e., each site is independent of all others. Therefore, no correlation effects are taken into account. This can be seen as the Gutzwiller approximates the ground state energy poorly at higher interaction (U/t) values.

5.3.1 Gutzwiller Acceptance Ratio

We want to calculate the probability of going from state n to state n' :

$$P(n \rightarrow n') = \min \left(1, \frac{|\psi_G(n')|^2}{|\psi_G(n)|^2} \right). \quad (15)$$

Let's now work out this ratio. We begin by using the relation we found earlier for $\langle n|\psi_G(n)\rangle$:

$$\frac{|\psi_G(n')|^2}{|\psi_G(n)|^2} = \frac{|\prod_{i=1}^M f_{n'_i}|^2}{|\prod_{j=1}^M f_{n_j}|^2}, \quad (16)$$

where n and n' differ only by $n'_i = n_i + 1$ and $n'_{i+1} = n_{i+1} - 1$. Therefore, we find

$$\frac{|\psi_G(n')|^2}{|\psi_G(n)|^2} = \frac{|f_{n_i+1} f_{n_{i+1}-1}|^2}{|f_{n_i} f_{n_{i+1}}|^2}. \quad (17)$$

5.3.2 Gutzwiller Local Kinetic Energy

Since the kinetic energy term is more complicated, below is a more in-depth calculation for a single term of a nearest-neighbor pair:

$$T_{\text{loc}}^i = \frac{\langle n | -tb_i^\dagger b_{i+1} - tb_{i+1}^\dagger b_i | \psi_G \rangle}{\langle n | \psi_G \rangle} \quad (18)$$

$$= -t \sum_{n'} \langle n | b_i^\dagger b_{i+1} + b_{i+1}^\dagger b_i | n' \rangle \frac{\psi_G(n')}{\psi_G(n)} \quad (19)$$

$$= -t \sum_{n'} \left[\sqrt{n'_{i+1}(n'_i + 1)} \langle n | n'_i + 1, n'_{i+1} - 1 \rangle \right] \frac{\psi_G(n')}{\psi_G(n)} \\ + \left[\sqrt{n'_i(n'_{i+1} + 1)} \langle n | n'_i - 1, n'_{i+1} + 1 \rangle \right] \frac{\psi_G(n')}{\psi_G(n)}. \quad (20)$$

From here, it is obvious that only surviving terms will be when $n'_{i+1} = n_{i+1} + 1$ and $n'_i = n_i - 1$ for the first term and $n'_{i+1} = n_{i+1} - 1$ and $n'_i = n_i + 1$ for the second term. Now, we will derive the simple form of the Gutzwiller projection onto our Fock state:

$$\langle n | \psi_G \rangle = \prod_{i=1}^M \langle n_i | \left(\sum_{m_i=0}^{n_{\text{max}}} \frac{f_{m_i} (b_i^\dagger)^{m_i}}{\sqrt{m_i!}} | 0_i \rangle \right) \quad (21)$$

$$= \prod_{i=1}^M \langle n_i | \left(\sum_{m_i=0}^{n_{\text{max}}} f_{m_i} | m_i \rangle \right), \quad (22)$$

since $(b_j^\dagger)^m | 0 \rangle = \sqrt{m!} | m \rangle$. We, therefore, find that

$$\langle n | \psi_G \rangle = \prod_{i=1}^M f_{n_i}. \quad (23)$$

Using this relation, the local kinetic energy for a single site is then

$$T_{\text{loc}}^i = -t \sqrt{n_{i+1}(n_i + 1)} \frac{f_{n_{i+1}} f_{n_{i+1}-1}}{f_{n_i} f_{n_{i+1}}} + \sqrt{n_i(n_{i+1} + 1)} \frac{f_{n_{i+1}+1} f_{n_i-1}}{f_{n_{i+1}} f_{n_i}}. \quad (24)$$

5.3.3 Gutzwiller Local Potential Energy

Since the potential term in the Hamiltonian is diagonal in the occupation basis, we can simply write is as

$$V_{\text{loc}}^i = \frac{U}{2} n_i (n_i - 1). \quad (25)$$

5.4 Jastrow

The general form of the Jastrow trial wavefunction is

$$|\psi_J\rangle = \exp \left(-\frac{1}{2} \sum_{i,j} \nu_{i,j} n_i n_j \right) |\Phi_0\rangle, \quad (26)$$

where

$$|\Phi_0\rangle = (b_{k=0}^\dagger)^N | 0 \rangle. \quad (27)$$

For a translationally invariant periodic lattice, the Jastrow potential depends only on the periodic Euclidean distance between sites. Let $s(i, j)$ be the shell index of the pair (i, j) , with equal distances assigned the same index, and let v_q be the potential for shell $q = 1, \dots, Q$, where Q is the number of distinct distance shells (do not confuse this with momentum indices). The shell set includes the on-site shell at distance zero. For a 1D ring of length L , the periodic distance is $\min(|i - j|, L - |i - j|)$. For a rectangular 2D lattice with dimensions $L_x \times L_y$, it is

$$r_{ij} = \sqrt{d_x^2 + d_y^2}, \quad d_x = \min(|x_i - x_j|, L_x - |x_i - x_j|), \quad d_y = \min(|y_i - y_j|, L_y - |y_i - y_j|). \quad (28)$$

There is one potential per distinct distance shell; the largest-distance potential is gauge-fixed to zero.

5.4.1 Acceptance Ratio

We want to calculate the probability of going from state n to state n' :

$$P(n \rightarrow n') = \min \left(1, \frac{|\psi_J(n')|^2}{|\psi_J(n)|^2} \right). \quad (29)$$

The condensate reference state has configuration amplitude proportional to $\prod_{i=1}^M (n_i!)^{-1/2}$. Thus, for a hop from site from to site to,

$$\frac{\Phi_0(n')}{\Phi_0(n)} = \sqrt{\frac{n_{\text{from}}}{n_{\text{to}} + 1}}. \quad (30)$$

The Jastrow factor and its derivatives with respect to the shell potentials are

$$\log \psi_J(n) - \log \Phi_0(n) = -\frac{1}{2} \sum_{i,j=1}^M v_{s(i,j)} n_i n_j, \quad (31)$$

$$O_q(n) = \frac{\partial \log \psi_J(n)}{\partial v_q} = -\frac{1}{2} \sum_{\substack{i,j=1 \\ s(i,j)=q}}^M n_i n_j. \quad (32)$$

Consequently, the change for a hop is

$$\Delta \log \psi_J = \sum_q v_q [O_q(n') - O_q(n)], \quad (33)$$

and the log acceptance ratio is

$$\log \left(\frac{|\psi_J(n')|^2}{|\psi_J(n)|^2} \right) = 2\Delta \log \psi_J + \log(n_{\text{from}}) - \log(n_{\text{to}} + 1). \quad (34)$$

In calculating $\Delta \log \psi_J$, only pairs involving either hopped site can contribute.

5.4.2 Jastrow Local Kinetic Energy

Similar to our energy calculations from the Gutzwiller trial wavefunction, we begin with the following form:

$$T_{\text{loc}}^i = \frac{\langle n | T | \psi_J \rangle}{\langle n | \psi_J \rangle} = \sum_{n'} \langle n | T | n' \rangle \frac{\langle n' | \psi_J \rangle}{\langle n | \psi_J \rangle}, \quad (35)$$

where the ratio on the right-hand side is simply the square root of the acceptance ratio we compute earlier. We can continue computing this explicitly to obtain the prefactors:

$$T_{\text{loc}}^i = -t \sum_{n'} \langle n | b_i^\dagger b_{i+1} + b_{i+1}^\dagger b_i | n' \rangle \frac{\langle n' | \psi_J \rangle}{\langle n | \psi_J \rangle} \quad (36)$$

$$\begin{aligned} &= -t \sum_{n'} \sqrt{n'_{i+1}(n'_i + 1)} \langle n | n'_{i+1} - 1, n'_i + 1 \rangle \frac{\langle n' | \psi_J \rangle}{\langle n | \psi_J \rangle} \\ &\quad - t \sum_{n'} \sqrt{n'_i(n'_{i+1} + 1)} \langle n | n'_i - 1, n'_{i+1} + 1 \rangle \frac{\langle n' | \psi_J \rangle}{\langle n | \psi_J \rangle}. \end{aligned} \quad (37)$$

From here, it is obvious that only terms where $n'_i + 1 = n_i$, $n'_{i+1} - 1 = n_{i+1}$ and $n'_i - 1$, $n'_{i+1} + 1 = n_{i+1}$ will survive for the left and right sums respectively. Our final expression is then

$$T_{\text{loc}}^i = -t \sqrt{(n_{i+1} + 1)n_i} \frac{\langle n_{i+1} + 1, n_i - 1 | \psi_J \rangle}{\langle n | \psi_J \rangle} - t \sqrt{(n_i + 1)n_{i+1}} \frac{\langle n_i + 1, n_{i+1} - 1 | \psi_J \rangle}{\langle n | \psi_J \rangle}. \quad (38)$$

6 VMC Algorithm

6.1 Markov Chain

A Markov chain is a sequence of random variables $X_1, X_2, \dots, X_n$ where the probability of being in a state X_{n+1} depends only on the previous state X_n . The Markov chain is defined by a transition matrix P where each element P_{ij} is the probability of transitioning from state i to state j . The transition matrix must satisfy the following properties:

1. $P_{ij} \geq 0$ for all i, j
2. $\sum_j P_{ij} = 1$ for all i

6.2 Metropolis–Hastings sampling

6.2.1 Detailed Balance

These Markov chains are built from two pieces: a proposal probability $g(n \rightarrow n')$ and an acceptance probability $A(n \rightarrow n')$. The proposal probability is the probability of proposing a new configuration n' given the current configuration n . The acceptance probability is the probability of accepting the proposed configuration. The product of these two pieces gives the transition probability from n to n' :

$$P(n \rightarrow n') = g(n \rightarrow n') A(n \rightarrow n'). \quad (39)$$

The transition probability must satisfy the detailed balance condition:

$$p(n) P(n \rightarrow n') = p(n') P(n' \rightarrow n), \quad (40)$$

where $p(n)$ is the probability of being in configuration n described by our sampling distribution. Since there is an infinite set of solutions to this equation, we pin the value of the backward move ($n' \rightarrow n$) to 1 and rearrange this equation to find the forward move ($n \rightarrow n'$) acceptance probability:

$$A(n \rightarrow n') = \frac{p(n') g(n' \rightarrow n)}{p(n) g(n \rightarrow n')}. \quad (41)$$

Since our proposals are symmetric, $g(n \rightarrow n') = g(n' \rightarrow n)$, we can simplify the acceptance ratio to

$$A(n \rightarrow n') = \frac{p(n')}{p(n)}. \quad (42)$$

In the case of variational Monte Carlo, we have $p(n) = |\psi_T(n)|^2$, so the acceptance ratio is

$$A(n \rightarrow n') = \frac{|\psi_T(n')|^2}{|\psi_T(n)|^2}. \quad (43)$$

6.3 Parameter optimization

We optimize the variational parameters of our trial wavefunction by performing gradient descent with stochastic reconfiguration on the energy expectation value. The gradient of the energy with respect to a variational parameter α is given by

$$f_k = -2 \left\langle \left(\hat{H} - E_\alpha \right) \hat{O}_k \right\rangle, \quad (44)$$

where $\hat{O}_k = \frac{\partial}{\partial \alpha_k} \log \psi_T$ and $E_\alpha = \langle \hat{H} \rangle$. The stochastic reconfiguration method is then used to update the variational parameters as follows:

$$\alpha_k^{(n+1)} = \alpha_k^{(n)} - \eta \sum_j S_{kj}^{-1} f_j, \quad (45)$$

where η is the learning rate and $S_{kj} = \langle \hat{O}_k \hat{O}_j \rangle - \langle \hat{O}_k \rangle \langle \hat{O}_j \rangle$ is the covariance matrix of the operators $\hat{O}_k$ and $\hat{O}_j$.

7 Code Structure

File	Responsibility
<code>lattice.jl</code>	Defines lattice and adjacency matrix
<code>MC_integration.jl</code>	Monte Carlo simulation
<code>moves.jl</code>	Metropolis–Hastings acceptance ratio
<code>measurements.jl</code>	Local energy, energy gradient, and $\Delta \log \psi$ calculations
<code>gradient_descent.jl</code>	Gradient descent with stochastic reconfiguration
<code>system.jl</code>	Stores all system parameters and lattice
<code>wavefunction.jl</code>	Stores wavefunction parameters
<code>VMCBoseHubbard.jl</code>	Module including all imports and exports
<code>main.jl</code>	Parse terminal input and run VMC

8 Benchmarks and Validation

Describe any checks against exact diagonalization, DMRG, or published results. Include convergence plots (energy vs. Monte Carlo steps, energy vs. variational parameter) as figures.

We have benchmarked our Gutzwiller and Jastrow trial wavefunctions by comparing ground-state energy VMC results with exact-diagonalization results for systems up to $M = 12$ sites and $N = 12$ particles. In addition, we confirmed the optimal Gutzwiller parameter for several interaction strengths by digitizing Werner Krauth’s Gutzwiller results from his paper [INCLUDE REFERENCE]. Below are the benchmarking graphs described above:

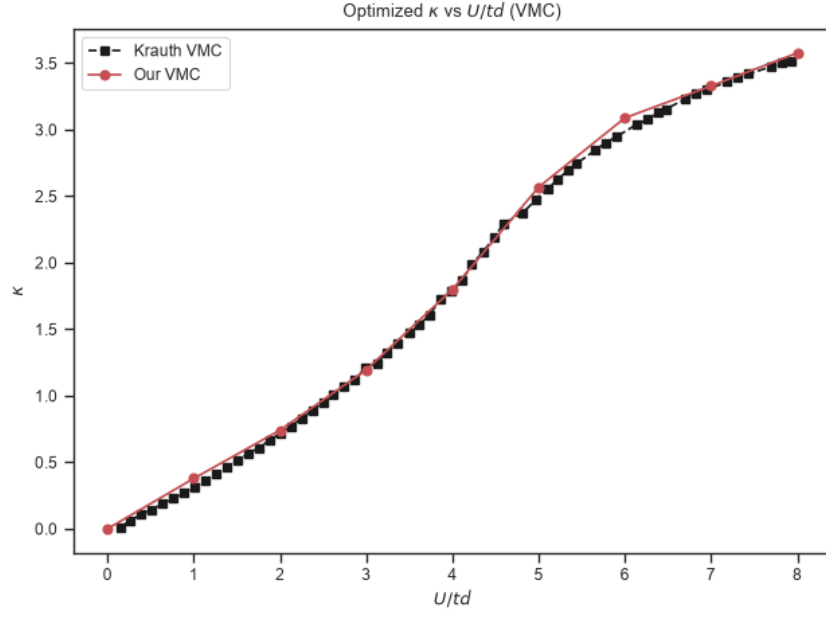


Figure 1:

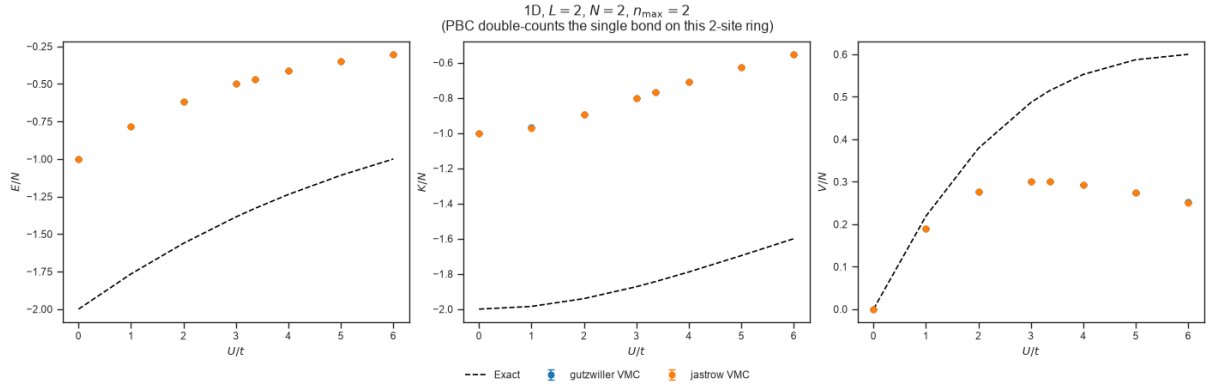


Figure 2:

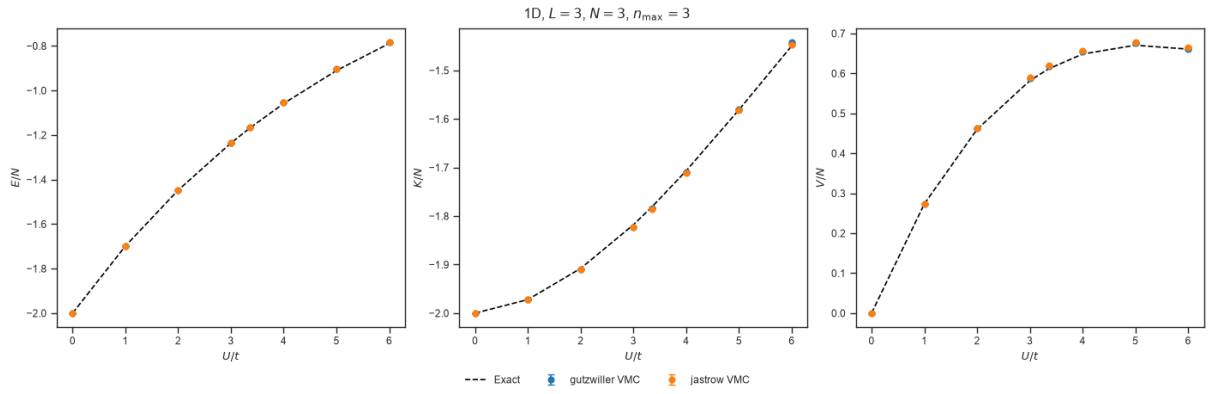


Figure 3:

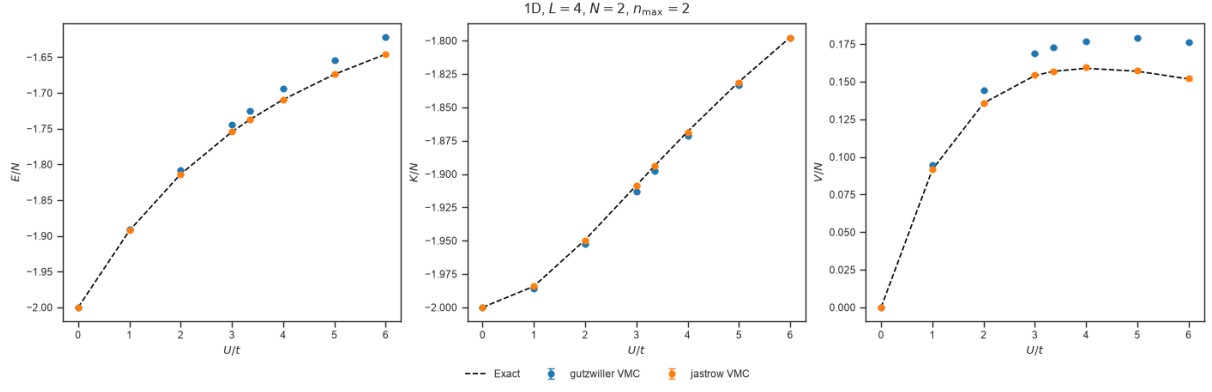


Figure 4:

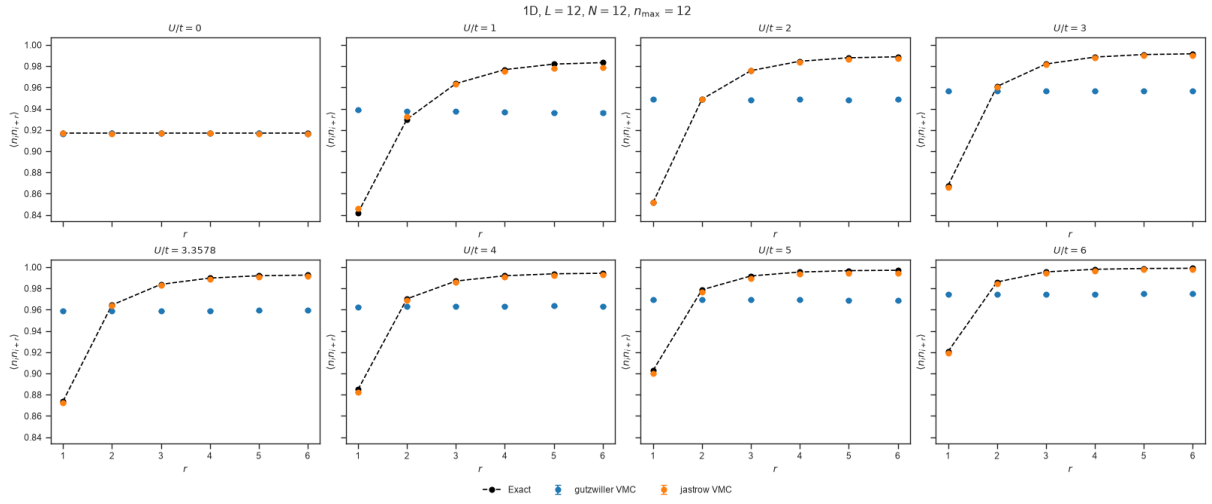


Figure 5:

9 Known Limitations and Future Work

- The Gutzwiller trial wavefunction does not take correlations between sites into account, causing it to fail in the high interaction (Mott insulating) regime

References